

DAIVIK BHATIA

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🐙 <https://gitlab.freedesktop.org/dtgs1208>

PROFESSIONAL SUMMARY

AI Engineer III by profession, and a Graphics Driver developer by passion. Currently achieving this by a full-time job at American Express and actively contributing to the Mesa 3D project with a focus on Broadcom (Raspberry Pi) drivers.

OPEN SOURCE CONTRIBUTIONS

Mesa

- Enabled `robustImageAccess2` for V3DV, while adding a custom NIR lowering pass to handle format swizzles for Out of Bound image loads.
- Investigated enabling of `OpTerminate` in broadcom drivers to align with the semantics of GLSL `discard`. Analysed the results from benchmarks (gfxbench) to assess the performance impact. There were no significant improvements observed by the usage of `discard` in place of `demote`.
- Enabled support for subgroup arithmetic operations and refined Compute Shader Dispatch supergroup packing logic.
- Improved the readability of code by replacing raw ints by V3DX based enums in both V3D and V3DV drivers.
- Other code refactoring and optimizations, including dropping unused methods and flags, improving barrier handling for secondary command buffers, investigation of enablement of `robustBufferAccess2` etc.
- Relevant Merge Requests [🔗](#)

Pyboy Emulator

07/2024

- Move the frame limiter function from OpenGL Window and SDL2 Window to PyBoy WindowPlugin. [Link](#) [🔗](#)
- Added support for GameShark API to allow users to use cheats in PyBoy. [Link](#) [🔗](#)

WORK HISTORY

AI Engineer III

08/2022 – Present

American Express

Gurgaon, India

- Made a 3 step solution to answer customer queries from client policy using LLMs like BERT, Roberta and GPT-J and using Flair-based ML models. Dockerized the solution and deployed it by building a flask-based API.
- Developed ML models for an entity matching project, enhancing data accuracy and reducing errors by integrating proprietary data with third-party vendor data. Utilized models such as Random Forest, KNN, XGBoost, and Logistic Regression.
- Led the development of a new hierarchy system for B2B clients, grouping clients together at Global, Regional, Market level using Machine Learning and Name Matching algorithms.

ML Engineer

Koo App

01/2022 – 08/2022

Bengaluru, India

- Designed and implemented a REST API utilizing Tesseract and EasyOCR for efficient text extraction, resulting in better categorisation.
- Developed an API for NSFW content detection using open source model, NSFW JS (Inception v3).
- Developed and Deployed an API for image captioning auto-generation using BLIP (Open source) for images without texts.
- Designed and implemented an end-to-end image classification pipeline utilizing machine learning models like Random Forest, KNN and XGBoost achieving better results.

R&D Engineer (AI/ML)

Logic Fruit Technologies

07/2021 – 01/2022

Gurgaon, India

- Participated in DAC SDC 2021 research competition which required training of an object detection model that had to be implemented on Avnet Ultrascale96 v2 FPGA board.
- Trained and Quantized models like efficientdet, yolov3, yolov4, yolov5, yolov3 tiny and yolov4 tiny across frameworks like tensorflow, pytorch, caffe, darknet. The final result was 0.81 IOU with 72 FPS.
- Trained an image classification model (Resnet50) using TensorFlow framework on a custom dataset and implemented it on Alveo U200 data acceleration card. Used Xilinx Vitis-AI for model quantization and compilation.

SKILLS

Graphics & Systems — C, C++, Vulkan, OpenGL, GLSL,

Development Tools — Git, GDB, VK-GL-CTS, Piglit,

General Programming — Python, SQL, Java script

EDUCATION

Bachelors of Technology

ADGIPS, GGSIPU

08/2017 – 06/2021 | Delhi, India

LANGUAGES

English

Hindi